

Exact Structural Reductions and Canonical Vortex Stretching Remainders for Periodic Three-Dimensional Navier-Stokes Flow

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Abstract

This article develops an exact reduction and falsification framework for the global-regularity problem for incompressible three-dimensional Navier-Stokes flow on the periodic torus. Starting from the projected Eulerian equation, we pass through the mild formulation, the volume-preserving flow map, the cofactor and Cauchy-Green variables, and a graph representation of vortex-stretching interactions. At each stage we distinguish identities that are unconditional for a smooth solution from estimates that would be required to continue that solution globally. The programme yields four principal results. First, the Lagrangian pullback and cofactor dynamics provide an information-complete geometric rewriting, but their ellipticity is controlled by the same accumulated strain that drives the original continuation problem. Second, explicit divergence-free families and singular-value examples show that volume preservation, one-dimensional streamline data, endpoint telescoping, and determinant constraints cannot by themselves control transverse deformation. Third, the cubic gradient remainder left by the enstrophy estimate is reduced exactly, modulo periodic divergences, to the mean vortex-stretching scalar. Fourth, a helical-triad compatibility graph represents the same obstruction, while single-triad saturation and weakly weighted incompatible cycles rule out a universal deficit derived from unweighted graph compatibility alone. The resulting synthesis is a conditional equivalence theorem and an obstruction-localization theorem: it does not establish global regularity, but identifies the precise uniform weighted estimate that a successful closure must supply. The framework is intended both as a rigorous account of the completed reductions and as a reusable test suite for proposed global estimates.

Keywords incompressible Navier-Stokes equations; global regularity; vortex stretching; Lagrangian flow; cofactor metric; Betchov identity; helical triads; obstruction analysis

1 Introduction

Let $(u=u(x,t))$ be a divergence-free velocity field on the three-dimensional torus $\mathbb{T}^3$, with pressure (p) , viscosity $\nu>0$, and smooth divergence-free initial data (u_0) . The equations are

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0.$$

The global existence and smoothness of solutions for arbitrary smooth periodic data remains one of the central open problems in mathematical fluid mechanics. Local strong solutions, global weak solutions, partial regularity, scale-critical continuation criteria, vorticity-direction criteria, and small-data results delimit the problem sharply, but none supplies a universal a priori estimate at the required critical level [1–8].

The purpose of this paper is not to introduce a new formal coordinate system and then treat that change of notation as a proof. Its purpose is to carry a long sequence of candidate reductions to their exact endpoint, to record which steps are reversible, and to falsify several tempting closures by explicit counterexamples. The central methodological rule is simple: an algebraic identity, a conditional equivalence, a finite-dimensional computation, and a uniform global estimate are different evidence classes. They must not be interchanged.

The source programme behind the paper explored four mutually connected routes: projected and mild Eulerian formulations; volume-preserving Lagrangian geometry; directional and graph coordinates; and Fourier-helical interaction graphs. The main gain from combining these routes is that their terminal obstructions agree. In physical space, the obstruction is the signed mean vortex-stretching scalar. In deformation variables, it is unbounded anisotropy of the cofactor metric. In Fourier-helical variables, it is the absence of a universal weighted phase deficit. These are not three solved subproblems; they are three exact representations of the same unresolved mechanism.

1.1 Contributions

The article makes the following bounded claims.

- 1 It provides an exact, reversible chain from the projected Eulerian equation to the mild equation and, for as long as a smooth flow map exists, to a cofactor-metric Lagrangian system.
- 2 It proves that determinant-one information cannot bound deformation anisotropy and that streamline derivatives cannot recover transverse gradients, using elementary explicit families.
- 3 It identifies the canonical scalar remainder of a broad divergence-reorganization of the cubic (H^1) energy term and reduces its spatial mean to the vortex-stretching integral.
- 4 It translates that scalar into a helical-triad interaction representation and proves two no-go statements for graph-only quantitative deficits.
- 5 It states a synthesis theorem that isolates the surviving proof obligation as a uniform, scale-compatible weighted estimate. The theorem is a reduction theorem, not a global-regularity theorem.

1.2 Proof-status discipline

The status terminology used throughout is summarized below. This table is part of the mathematical statement of the paper, not administrative metadata.

Status	Meaning in this article	Does it prove global regularity
Exact identity	Equality follows for every sufficiently smooth solution on its lifespan	No
Conditional equivalence	Two descriptions are equivalent while the required smooth diffeomorphism or convergence exists	No
Counterexample or no-go result	A proposed class of closure mechanisms is ruled out	No
Finite or truncated verification	An identity is checked in a Galerkin or graph truncation	No
Required closure estimate	A uniform bound sufficient for continuation, not established here	Would contribute if proved

2 Setting and analytical background

We work on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ and normalize integrals only when explicitly stated. Spatial indices are summed. The velocity gradient is $(M = \nabla u)$, its symmetric and antisymmetric parts are

$$S = \frac{1}{2}(M + M^T), \quad \Omega = \frac{1}{2}(M - M^T),$$

and the vorticity is $\omega = \nabla \times u$. The Leray projector onto divergence-free vector fields is denoted by $\mathbb{P}$. Applying $\mathbb{P}$ eliminates pressure and gives

$$\partial_t u - \nu \Delta u = -\mathbb{P} \nabla \cdot (u \otimes u).$$

For a strong solution this is equivalent to the mild equation

$$u(t) = e^{\nu t \Delta} u_0 - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (u \otimes u)(s) ds.$$

The energy identity is

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2.$$

This estimate is global at the energy level but supercritical with respect to the natural three-dimensional scaling. Differentiating once, multiplying by ∇u , and integrating yields an (H^1) -level balance containing a signed cubic term. The inability to control that term uniformly at the critical scale is the recurring endpoint of the reductions below.

Local well-posedness can be obtained in several classical strong spaces, including scale-critical settings [4,5]. Leray-Hopf weak solutions exist globally [2], and their possible singular set is constrained by partial regularity [6]. Prodi-Serrin conditions and geometric criteria based on vorticity direction give sufficient regularity under additional integrability or coherence assumptions [7,8]. Our claims are compatible with these results: we neither weaken their hypotheses nor infer that their conditions hold for arbitrary smooth data.

3 From coordinate proposals to an exact audit chain

The initial coordinate programme began with a relative variable carrying a value of (u) at a selected point. Such a coordinate is exact at the instant of definition, but it contains no evolution law. Successive spatial derivatives add local jet information; eliminating pressure with $\mathbb{P}$ adds the correct nonlocal projection. These operations produce a valid tangent description, yet none controls the future smoothness of the solution.

This distinction is easiest to express as an ablation principle. Suppose a proposed graph or coordinate system is claimed to be sufficient for global continuation. Remove, one at a time, the assumptions that the solution is already smooth, that the graph remains invertible, that a limiting Picard sequence is Cauchy, or that a critical norm stays finite. If removing one assumption destroys the argument and the assumption is equivalent to the desired conclusion, the coordinate system has not closed the problem. It has restated it.

Finite Galerkin systems are useful audit objects because all operations are classical ordinary differential equations. They verify signs, symmetries, conservation laws, and graph bookkeeping. They do not, without bounds uniform in the truncation scale, pass to a smooth infinite-dimensional limit. Likewise, an exact residual verifier can establish that a candidate function solves the equation, but it does not construct that function or prove that an iterative sequence converges globally.

The exact chain retained in this paper is therefore

$$\text{“projected PDE”} \leftrightarrow \text{“mild equation”} \leftrightarrow \text{“flow-map system”} \leftrightarrow \text{“cofactor metric”} \leftrightarrow \text{“canonical stretching remainder”} \leftrightarrow \text{“weighted triad graph”},$$

where the middle equivalences are conditional on smooth invertibility and the final arrow is a representation of the nonlinear interaction rather than an estimate. The construction is valuable because every route exposes the same missing quantitative input.

4 Mild fixed points and the Cauchy gate

Define the bilinear Duhamel operator

$$B(u,v)(t) = -\int_0^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (u \otimes v)(s) ds.$$

Then $(u = e^{\nu t \Delta} u_0 + B(u, u))$. A Picard sequence $(u^{\wedge \{n+1\}} = e^{\nu t \Delta} u_0 + B(u^{\wedge \{n\}}, u^{\wedge \{n\}}))$ is standard locally in time when the chosen norm makes (B) contractive. Extending this construction globally requires a uniform bound that prevents the contraction interval from collapsing or allows continuation across successive intervals.

Proposition 4.1 Conditional Picard limit. Let (X_T) be a Banach space in which the heat flow is bounded and (B) satisfies the estimates required for a local contraction. If the Picard sequence is Cauchy in (X_T) , its limit satisfies the mild equation on $([0, T])$. This statement supplies no bound ensuring that the sequence is Cauchy for arbitrary (T) .

Proof. Bilinearity and continuity give $(B(u^{\wedge \{n\}}, u^{\wedge \{n\}}) \rightarrow B(u, u))$ whenever $(u^{\wedge \{n\}} \rightarrow u)$ in (X_T) . Passing to the limit in the iteration proves the mild equation. The argument begins with Cauchy convergence and therefore cannot be used to prove it. $\square$

The Fourier integrating factor gives the same conclusion. For $(k \neq 0)$,

$$\hat{u}_k(t) = e^{-\nu |k|^2 t} \hat{u}_k(0) - \int_0^t e^{-\nu |k|^2(t-s)} \{\mathbb{P} \nabla \cdot (u \otimes u)\}_k(s) ds.$$

The linear dissipation is diagonal, but the quadratic convolution remains. Repeated integration by parts reorganizes derivatives and reveals cancellations in scalar energies; it does not remove the convolution or provide an absolute estimate. Taylor expansion has the same logical boundary: truncating the expansion leaves a remainder whose control is precisely the analytic question.

5 Volume-preserving Lagrangian geometry

Let $\Phi(a, t)$ solve

$$\partial_t \Phi(a, t) = u(\Phi(a, t), t), \quad \Phi(a, 0) = a,$$

and set $(A = D_a \Phi)$, $(J = \det A)$, and $(C = A^{\wedge \{-T\}})$ when (A) is invertible. Incompressibility gives $(J = 1)$. The deformation equation is

$$\partial_t A = (\nabla u \circ \Phi) A.$$

The cofactor metric is

$$K = C^T C = A^{-1} A^{-T}.$$

Because $(J=1)$, $\det K=1$. Differentiating $(C=A^{\wedge\{-T\}})$ gives an exact evolution law. With the source convention in which (S) is pulled back along Φ ,

$$\partial_t K = -2C^T T S C.$$

The precise matrix placement follows from the convention for (A) ; the invariant content is that the metric growth is driven by strain.

5.1 Pulled-back equation

Write $(v(a,t)=u(\Phi(a,t),t)=\partial_t \Phi(a,t))$ and $(q(a,t)=p(\Phi(a,t),t))$. The material derivative becomes ∂_t at fixed (a) . The pressure gradient transforms by $(C \nabla_a q)$, and the Laplacian becomes a variable-coefficient elliptic operator with coefficient (K) . Thus, while Φ remains a smooth volume-preserving diffeomorphism,

$$\partial_{tt} \Phi + C \nabla_a q = \nu \operatorname{Div}_a (K \nabla_a \partial_t \Phi).$$

This formula is an exact geometric rewriting. It does not make the ellipticity uniform: the eigenvalues of (K) are reciprocals of squared singular values of (A) .

Proposition 5.1 Conditional Eulerian-Lagrangian equivalence. On every interval on which Φ is a smooth volume-preserving diffeomorphism with bounded (A) and $(A^{\wedge\{-1\}})$, the Eulerian Navier-Stokes system and the pulled-back cofactor system are equivalent. Global solvability of one formulation follows from the other only if these geometric bounds persist globally.

Proof. The forward direction is the chain rule, the Piola identity, and change of variables. For the reverse direction define $(u(x,t)=\partial_t \Phi(\Phi^{\wedge\{-1\}}(x,t),t))$ and $(p(x,t)=q(\Phi^{\wedge\{-1\}}(x,t),t))$. Invertibility and the same identities recover the Eulerian equations. Loss of invertibility or ellipticity invalidates the reverse construction. $\square$

5.2 Determinant one does not control condition number

Let the singular values of (A) be $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$. Volume preservation gives $\sigma_1 \sigma_2 \sigma_3 = 1$, but permits σ_1/σ_3 to be arbitrarily large. The diagonal family

$$A_\lambda = \operatorname{diag}(\lambda, \lambda^{-1}, 1), \lambda \rightarrow \infty,$$

has determinant one and unbounded condition number. Hence determinant control alone supplies no uniform ellipticity for (K) .

The logarithmic singular-value growth obeys a directional strain law. At differentiability points of the singular values,

$$|d(-\log \sigma_i)/dt| \leq \|S(\cdot, t)\|_{L^\infty},$$

and therefore

$$\kappa(A(t)) \leq \kappa(A(0)) \exp(2 \int_0^t \|S(s)\|_{L^\infty} ds).$$

The bound is conditional on the accumulated (L^∞) strain. It relocates the continuation criterion but does not establish the required integral estimate.

6 Directional coordinates and explicit information loss

A natural attempt is to follow a streamline or a material curve and express the change of speed as a one-dimensional ratio. If $(q=|u|)$ and $(e=u/q)$ on the set where $(q>0)$, incompressibility implies

$$0 = \nabla \cdot (qe) = e \cdot \nabla q + q \nabla \cdot e,$$

so along arclength (s) ,

$$dq/ds = -q \nabla \cdot e.$$

This is exact but incomplete: the evolution of (e) involves transverse derivatives and pressure. Moreover, the coordinate degenerates where $(u=0)$.

6.1 A counterexample family

Proposition 6.1 Streamline derivative does not determine the transverse gradient. There are smooth periodic divergence-free fields for which the derivative of (u) along its own direction vanishes identically while $\|\nabla u\|$ can be arbitrarily large.

Proof. Let

$$u_\theta(x, y, z) = (\cos\theta(z), \sin\theta(z), 0)$$

for any smooth periodic θ . Then $\nabla \cdot u = 0$, $-|u| = 1$, and

$$u \cdot \nabla = \cos\theta(z) \partial_x + \sin\theta(z) \partial_y.$$

Because (u_θ) depends only on (z) , $((u \cdot \nabla)u = 0)$. However,

$$\partial_{zu} \theta = \theta'(z) (-\sin\theta(z), \cos\theta(z), 0),$$

so $\|\nabla u\|$ contains $\|\theta'\|$, which may be arbitrarily large. Thus a one-dimensional derivative along streamlines loses transverse information. $\square$

An evolving separation frame repairs the information loss only by restoring the full deformation matrix. If $\delta x = A \delta a$, then $\partial_t A = (\nabla u \circ \Phi)A$ and, while (A) is invertible,

$$\nabla u \circ \Phi = (\partial_{tA})A^{-1}.$$

This is information-complete but not estimate-complete. Second variations encode the Hessian, again without a uniform bound.

6.2 Telescoping is not control

Partitioning a path and multiplying local speed ratios produces

$$\prod_j q(s_{j+1})/q(s_j) = q(s_N)/q(s_0).$$

Taking logarithms gives a telescoping sum. The equality converts an accumulation into endpoint data but supplies no a priori bound for the endpoint ratio. A material path need not close, and even a closed spatial loop at one time does not neutralize time-dependent deformation. Treating telescoping as a bound is therefore a logical error.

7 The cubic gradient remainder

Differentiating the equation, testing by ∇u , and integrating over $\mathbb{T}^3$ gives

$$\frac{1}{2} d\|\nabla u\|_2^2/dt + \nu \|\Delta u\|_2^2 = -I(u),$$

with a convention-dependent sign and

$$I(u) = \int_{\mathbb{T}^3} \partial_k u_j \partial_j u_i \partial_k u_i \, dx.$$

Integration by parts can move pieces of the pointwise cubic density into a divergence, but on the torus every divergence has zero spatial mean. Thus the zero Fourier mode of the cubic density survives every periodic divergence reorganization.

7.1 Canonical zero-mode statement

Let $(g(x) = \partial_k u_j \partial_j u_i \partial_k u_i)$. Write $\langle g \rangle$ for its spatial mean. Then a periodic vector field (F) can change (g) to $(g - \nabla \cdot F)$, but cannot change $\langle g \rangle$. Conversely, any two periodic densities differing by a divergence have the same mean. The quotient of scalar densities by periodic divergences is therefore represented at the scalar level by the zero mode.

Theorem 7.1 Canonical scalar remainder. For every smooth periodic divergence-free velocity field, the scalar obstruction left by any exact periodic divergence reorganization of the (H^1) cubic density is its mean $\langle I(u) \rangle$. With the conventions above,

$$I(u) = - \int_{\mathbb{T}^3} \omega \cdot S \omega \, dx.$$

The identity is exact; it is not a sign estimate.

Proof. In matrix notation the density is $(g = \text{tr}(M^T M))$. Decompose $(M = S + \Omega)$. Symmetry and antisymmetry eliminate traces containing one symmetric factor and one antisymmetric factor in the relevant cyclic pairings, yielding

$$\text{tr}(M^T M) = \text{tr}(S^T S) - \frac{1}{4} \omega^T S \omega.$$

For smooth periodic incompressible fields, the Betchov identity gives

$$\int_{\mathbb{T}^3} \text{tr}(S^3) \, dx = -\frac{3}{4} \int_{\mathbb{T}^3} \omega^\wedge \text{TS} \omega \, dx.$$

Combining the two relations proves $(I = -\int \omega^\wedge \text{TS} \omega)$. The invariance of the mean under addition of a divergence proves canonicity in the stated quotient. $\square$

This theorem explains why repeated spatial integration by parts reaches a stable endpoint. The remaining mean is the enstrophy-production term. Because $\omega \cdot S\omega$ has no fixed sign, trace-free strain alone does not control it.

8 Strain eigengeometry and feedback

Let $(S e_i = \lambda_i e_i)$, $\lambda_1 + \lambda_2 + \lambda_3 = 0$, and write $\omega = \sum \omega_i e_i$. Then

$$\omega \cdot S\omega = \sum_{i=1}^3 \lambda_i |\omega_i|^2.$$

The zero trace allows extension in one direction to be balanced by compression in others, but the vorticity may preferentially align with an extensional eigenvector. Consequently, the missing information is a correlation between strain eigenvalues and vorticity direction, not merely a scalar bound on $(\text{tr} S)$.

Geometric depletion criteria in the literature exploit coherence of vorticity direction or related directional structure [8]. The present reduction is consistent with that philosophy but does not prove the required coherence for arbitrary data. The angular factor appearing in the Biot-Savart representation provides cancellation when directions align; its absolute value can be bounded schematically by a multiple of $|\sin \theta|$. The unresolved step is a feedback mechanism forcing such alignment or another compensating structure at all dangerous scales.

Define $(r = \omega^\wedge \text{TS} \omega)$. The vorticity equation is

$$D_t \omega = S\omega + \nu \Delta \omega.$$

Differentiating (r) along trajectories produces terms of the form

$$D_{\text{tr}} = \omega^\wedge \text{TS}^2 \omega - \omega^\wedge T(\nabla^2 p) \omega + \text{“viscous derivative terms”} + \text{“additional symmetric-gradient terms”}.$$

The pressure Hessian is nonlocal and the viscous terms are not an autonomous scalar function of (r) . Thus (r) is a useful graph node but not a closed scalar dynamical variable.

9 Critical scale and the heat-weighted flux barrier

Under the Navier-Stokes scaling $(u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t))$, the homogeneous $\dot{X}^{1/2}$ norm is invariant. A dyadic decomposition with Littlewood-Paley blocks (u_j) localizes nonlinear transfer across shells. The critical energy schematically satisfies

$$E_c(t) \sim \sum_j 2^{2j} \|u_j(t)\|_2^2.$$

The mild formulation introduces heat weights $(e^{-\{c\nu 2^{2j}\}(t-s)})$. These weights strongly suppress frequencies satisfying $\nu 2^{2j}(t-s) \ll 1$, but are order one in the boundary layer

$$\nu 2^{2j}(t-s) = O(1).$$

The dangerous interactions therefore concentrate near comparable scales and near the terminal time. Crude absolute-value estimates at this boundary lose the signed cancellation needed at criticality. The resulting (K^0) barrier is not a divergence of a power of the cutoff; it is the failure to obtain a vanishing or summable gain uniformly at the cutoff.

This observation narrows, but does not solve, the problem. A sufficient estimate must exploit signed flux, angular depletion, phase incompatibility, viscosity, or a combination that is uniform in scale and time. An estimate proved only for a fixed truncation or separated scales does not pass this gate.

10 Helical triads and a compatibility graph

For each nonzero Fourier mode (k) , choose helical basis vectors $(h_k^\pm)$ satisfying

$$ik \times h_k^\pm = s|k| h_k^\pm, \quad s \in \{+1, -1\},$$

and expand $\hat{u}(k) = \sum_s u_k^\pm h_k^\pm$. The quadratic nonlinearity couples triads $(k+p+q=0)$, with coefficients determined by wave-vector geometry and helicity signs [10]. The vortex-stretching mean can therefore be represented as a signed sum of weighted triad interactions.

Construct a graph whose vertices encode active helical modes or interaction states and whose hyperedges encode admissible triads. Edge phases record the phase combination entering the real cubic transfer. Shared modes impose compatibility constraints between incident triads. The graph representation is exact once the Fourier field and basis convention are fixed.

The representation suggests a possible cancellation mechanism: incompatible phase preferences around cycles could prevent all edge contributions from attaining their individual maximal sign simultaneously. This idea is real but insufficient without weights.

10.1 Single-triad saturation

Proposition 10.1 No universal deficit from compatibility alone. A graph class that includes a single active triad admits configurations attaining the edgewise phase optimum. Therefore no strictly positive universal deficit from the sum of absolute edge weights follows from compatibility alone.

Proof. For one triad there is one relevant phase combination. The phases of the three complex amplitudes can be chosen so that the real cubic interaction has its maximizing sign. There is no cycle constraint to create frustration. Hence any proposed estimate of the form

$$|\sum_e w_e \Psi_e| \leq (1-\delta) \sum_e |w_e|, \delta > 0,$$

fails uniformly on a class permitting a single nonzero edge. $\square$

10.2 Weakly weighted incompatible cycles

Proposition 10.2 Unweighted incompatibility does not imply a weighted gap. Even if a graph contains an incompatible cycle, the weights on the frustrating edges may tend to zero. Thus a graph-theoretic incompatibility statement without a quantitative lower bound on the relevant weights cannot yield a uniform positive deficit.

Proof. Start with edges whose phases can nearly maximize the dominant weighted sum. Add an incompatible cycle with total weight ε . The cycle makes the unweighted constraint system inconsistent, but the loss in the weighted optimum is at most $(O(\varepsilon))$. Sending $\varepsilon \rightarrow 0$ eliminates any graph-independent positive gap. $\square$

The surviving target must therefore be a *weighted* statement tied to Navier–Stokes amplitudes, heat factors, and geometry. Possible routes include a lower bound on the weight carried by frustrating cycles, a viscosity-dependent compensation of near-saturating trees, or an analytic exclusion of isolated coherent triads at the critical boundary. None is proved here.

11 Main synthesis theorem

The preceding sections can be assembled into a single theorem that states exactly what the reduction accomplishes.

Theorem 11.1 Structural equivalence and obstruction localization. Let (u) be a smooth periodic solution of the three-dimensional incompressible Navier–Stokes equations on $([0, T])$. Then: (i) the projected, mild, and Eulerian differential formulations are equivalent; (ii) while the flow map remains a smooth diffeomorphism, the Eulerian system is equivalent to the volume-preserving cofactor-metric system; (iii) the (H^1) cubic scalar remainder is, modulo periodic divergences, the vortex-stretching mean $(-f\omega \cdot S\omega)$; and (iv) this mean admits a weighted helical-triad graph representation. None of volume preservation, streamline data, endpoint telescoping, unweighted phase compatibility, or a fixed Galerkin verification supplies a uniform bound for this remainder. Consequently, this chain yields global regularity only if supplemented by a scale-uniform weighted estimate controlling the canonical remainder or an equivalent critical flux.

Proof. Item (i) follows from applying the Leray projection and Duhamel’s formula, with the standard regularity needed to differentiate or integrate the equation. Item (ii) is Proposition 5.1. Item (iii) is Theorem 7.1. Item (iv) follows from the helical Fourier decomposition and triadic convolution. The failure statements follow respectively from $(A_\lambda = \text{diag}(\lambda, \lambda^{-1}, 1))$, Proposition 6.1, the endpoint identity of Section 6.2, Propositions 10.1–10.2, and the absence of truncation-uniform estimates. The final sentence is logical: every exact representation still contains the same uncontrolled scalar or its equivalent. $\square$

11.1 A sufficient closure target

The theorem suggests a family of genuinely sufficient targets. One example would be a bound of the schematic form

$$\int_0^T |\int_{\mathbb{T}^3} \omega \cdot S\omega \, dx| \, dt \leq \Phi(u_0, \nu, T),$$

where Φ is finite for every smooth divergence-free datum and is strong enough, together with the enstrophy balance, to prevent blow-up. Another would be a dyadic estimate with a positive summable gain at the heat boundary, uniform in the ultraviolet cutoff. A third would be a weighted graph deficit whose constant is stable under mode activation and degeneration of small weights. Merely writing any one of these inequalities is not a proof; the analytic derivation is the remaining problem.

11.2 Why the result is still useful

Obstruction localization has two uses. First, it prevents repeated investment in routes that fail for an elementary reason, such as determinant-only control or unweighted phase frustration. Second, it gives proposed proofs a short adversarial test suite. A candidate estimate should be tested against the shear-rotation family (u_θ) , anisotropic determinant-one matrices, isolated helical triads, weakly weighted incompatible cycles, and the terminal heat boundary. Failure on any test reveals the missing hypothesis before a long argument is built on it.

12 Relationship to established results

The global problem and its acceptable formulations are stated by Fefferman for the Clay Mathematics Institute [1]. Leray's weak-solution construction [2] and later partial-regularity theory [6] show that global distributional existence is distinct from global smoothness and uniqueness. The mild formulation and critical-space theory follow the classical Fujita-Kato and Kato frameworks [4,5]. Prodi-Serrin regularity criteria [7] show that additional spacetime integrability closes the problem conditionally.

The reduction of the cubic gradient term to vortex stretching is part of the classical strain-vorticity structure and is related to Betchov's identity [9]. It is not presented here as a newly discovered fluid identity. The contribution is the proof-status role assigned to it: it is the invariant zero-mode remainder shared by the explored divergence and graph reorganizations.

Constantin and Fefferman established regularity conditions based on the direction of vorticity [8]. Those results demonstrate that geometric depletion can be analytically decisive under quantitative hypotheses. The present angular and eigenframe analysis does not derive those hypotheses universally; instead, it explains where an unconditional version would have to enter.

Waleffe's helical decomposition [10] provides the natural triadic coordinates used in Section 10. Our compatibility graph is a bookkeeping and falsification layer over that representation. Its no-go propositions do not contradict triadic cancellation results. They show only that topology without quantitative weights cannot force the uniform deficit needed here.

13 Limitations and open theorem

The present manuscript has three deliberate limitations.

First, all equivalences involving Φ are conditional on a smooth invertible flow map. The paper does not infer persistence of that property from $\det D\Phi=1$. Second, the canonical remainder theorem identifies the scalar obstruction but gives no sign or universal magnitude bound. Third, the graph analysis excludes an unweighted universal deficit but does not classify every possible weighted analytic inequality.

Accordingly, the open theorem after this work can be stated compactly:

Open weighted closure problem. Prove, for every smooth divergence-free periodic datum, a viscosity- and data-dependent estimate that controls the heat-localized weighted helical-triad representation of $\int \omega \cdot S\omega$ uniformly in frequency cutoff and remains effective in the comparable-scale terminal layer $\nu K^2(t-s)=O(1)$; or prove an equivalent uniform estimate in the cofactor-metric or physical-space formulation.

This statement is intentionally stronger than compatibility of a fixed graph, exactness of a residual, or convergence of a fixed finite truncation. It includes the quantifiers that a global

proof must discharge.

14 Conclusion

The reduction programme reaches a clear endpoint. Projected, mild, Lagrangian, cofactor, vortex-stretching, and helical-graph descriptions can be connected exactly on the lifespan of a smooth solution. Their apparent differences do not remove the central difficulty: anisotropic strain can grow while volume is preserved; transverse gradients escape streamline data; periodic divergences cannot eliminate the zero mode; and graph incompatibility without quantitative weights cannot produce a uniform cancellation gap.

The main mathematical product is therefore a structural equivalence and no-go framework. It converts a diffuse collection of candidate mechanisms into one sharply delimited target: a uniform weighted control of the canonical vortex-stretching remainder at the critical heat boundary. Establishing that estimate remains open. Any future claim of global regularity based on this programme should be judged at that gate.

Appendix A Source-chain reconciliation

The research archive contains a physical sequence of thirty-nine PURE_MATH documents representing logical items 01–42. The numbering difference is caused by the combined file for items 12–15. The chain is grouped below by mathematical role.

Logical items	Role in the combined manuscript	Final status
01–03	Relative coordinate, spatial variation, pressure elimination	Exact local identities; no future bound
04–08	Finite graph, ablation, mild and energy formulations	Finite or conditional verification
09–11	Lifted moments, residual verifier, integral fixed point	Exact representation; construction open
12–15	Picard iteration, Cauchy passage, limit equation	Cauchy gate open globally
16–20	Fourier integrating factor, integration by parts, Taylor and heat fixed points	Nonlinearity survives; local theory only
21–25	Flow map, metric, cofactor, pulled-back PDE, geometric equivalence	Conditional on smooth invertibility
26–28	Metric evolution, chain ratios, finite arclength conversion	Exact identities; degeneracy remains
29–34	Speed-direction split, transverse counterexample, separation frame, singular values, telescoping, divergence decomposition	Several closure routes falsified
35–39	Canonical mean, Betchov reduction, eigengeometry, uniqueness modulo divergences, feedback graph	Exact obstruction localized
40–42	Helical phase graph, deficit test, split/recombination identity	Representation exact; universal graph deficit fails

The earlier standalone Papers 7–10 form conceptual prehistory. Paper 7 studies fixed and dynamic graph/null coordinates; Paper 8 isolates a critical shell-flux barrier; Paper 9 develops signed core/outer strain geometry; Paper 10 reduces a proposed global statement to explicit proof obligations. Their reusable content is incorporated in Sections 3, 8, 9, and 11 rather than treated as four consecutive proof chapters.

Appendix B Claim ledger

Claim	Evidence class	Dependency	Unresolved boundary
Pressure-free projected equation	Exact identity	Smooth divergence-free field	None at identity level
Mild formulation	Conditional equivalence	Strong solution or justified Duhamel passage	Global norm bound
Picard limit solves equation	Conditional implication	Cauchy convergence	Uniform global Cauchy estimate
$\det D\Phi=1$	Exact identity	Smooth flow map	Does not bound anisotropy
Cofactor-metric pullback	Conditional equivalence	Smooth invertible Φ	Uniform ellipticity
Streamline data are insufficient	Explicit counterexample	Family (u_θ)	Full transverse control still needed
Singular-value ratios are uncontrolled by determinant	Explicit counterexample	(A_λ)	Accumulated strain bound
Cubic remainder equals negative stretching mean	Exact identity	Smooth periodic incompressible field	No sign control
Stretching mean has helical-triad representation	Exact representation	Fourier/helical expansion	Weighted summability and cancellation
Universal unweighted graph deficit	Falsified	Single triad and small frustrating weights	Must include analytic weights
Global regularity	Not established	Would require a uniform closure estimate	Open

Appendix C Adversarial test suite for future estimates

A proposed global estimate arising from the programme should pass all of the following tests.

- 1 **Anisotropic volume test.** The estimate must remain informative for determinant-one deformation matrices with arbitrarily large condition number.
- 2 **Transverse-information test.** It must detect the gradients of $(u_\theta = (\cos\theta(z), \sin\theta(z), 0))$ even though the streamline derivative vanishes.
- 3 **Zero-mode test.** It must control the spatial mean of the cubic density, not only rewrite nonzero modes as divergences.
- 4 **Single-triad test.** It must permit saturation by one triad or explain analytically why the Navier-Stokes evolution cannot maintain that configuration at the dangerous boundary.
- 5 **Small-weight frustration test.** A graph deficit must be weighted and stable when incompatible edges carry arbitrarily small amplitude.
- 6 **Ultraviolet uniformity test.** Constants must be independent of Galerkin cutoff or graph size.
- 7 **Terminal heat-layer test.** The estimate must retain a summable gain when $\nu K^2(t-s) = O(1)$, where heat damping is not small.
- 8 **Continuation test.** The final inequality must imply finiteness of a recognized continuation quantity rather than assume that quantity in an intermediate coordinate transformation.

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